Ozgen Ozturk

University of Oxford
Department of Economics
10 Manor Rd, Oxford, OX1 3UQ
United Kingdom

Website: www.ozgenozturk.com
Email: Ozgen.Ozturk@economics.ox.ac.uk

Professional Experience

Sep 2023 - *Present*	***Postdoctoral Research Fellow*, University of Oxford**, Department of Economics
October 2023 - *Present*	***Academic Consultant*, Bank of England**
July 2022 - July 2023	***Economist*, Bank of England**, Monetary Analysis Directorate

Research Interests

Macrofinance, Firm Dynamics, Monetary Policy, Corporate Finance

References

Russell Cooper
European University Institute
Via delle Fontanelle 18
50014 Fiesole, Italy
Russell.Cooper@eui.eu

Federica Romei
University of Oxford
Department of Economics
10 Manor Rd, OX1 3UQ Oxford, United Kingdom
Federica.Romei@economics.ox.ac.uk

Árpád Ábrahám
University of Bristol
The Priory Road Complex
Priory Road, Clifton
BS8 1TU, Bristol, United Kingdom
Arpad.Abraham@bristol.ac.uk

Ambrogio Cesa-Bianchi
Bank of England
Threadneedle Street
London EC2R 8AH, United Kingdom
Ambrogio.Cesa-Bianchi@bankofengland.co.uk

Education

2017 - 2023	***PhD in Economics,* European University Institute**, Florence, Italy Advisors: Russell Cooper & Árpád Ábrahám
2017 - 2018	***MRes in Economics,* European University Institute**, Florence, Italy
2014 - 2017	***PhD studies in Economics - Course work and Qualifiers,* Middle East Technical University**, Ankara, Turkey
2012 - 2014	***MSc in Economics and Finance,* Middle East Technical University**, Ankara, Turkey
2007 - 2011	***BSc in Electrical and Electronics Engineering* Hacettepe University**, Ankara, Turkey

Publications

"Firm Inflation Uncertainty" (with *Nick Bloom, Phil Bunn, Paul Mizen, Gregory Thwaites, Ivan Yotzov*)

AEA Papers and Proceedings, Volume 113, May 2023 [NBER]

Working Papers

"Innovation, Financial Frictions, and Hysteresis Effects of Monetary Policy" (with *Aydan Dogan*)
This paper examines the link between access to external finance and the long-term impact of monetary policy on productivity growth. By leveraging loan-level data merged with firm-level balance sheet information, we show that firms' R&D expenditures decline after a monetary tightening, with heterogeneous responses. Firms that lack access to external finance for funding R&D activities experience sharper cuts in R&D spending compared to those with better access. Within an endogenous growth model with nominal rigidities and financial frictions, we

interpret this pattern as access to external finance enables firms to sustain innovation during periods of monetary tightening. Our model findings suggest that these short-term impact of monetary policy on R&D investment can have long-lasting effects on productivity, as current R&D efforts drive future productivity growth. Additionally, we show that when firms are provided with the financial flexibility to borrow to finance innovation activities, and monetary policy targets the output gap, it is possible to stabilise output without inducing hysteresis effects.

"TFPR: Dispersion and Cyclicality" (with *Russell Cooper*)
Conditionally Accepted at AEJ: Macro **[NBER]**
This paper studies the determinants of the dispersion and cyclicality of TFPR, a revenue based measure of total factor productivity. Recent business cycle models are built upon the assumption of countercyclical dispersion in TFPQ, a quantity based measure of total factor productivity, based on evidence of countercyclical dispersion in TFPR. But, these are very different measures of productivity. The distribution of TFPR is endogenous, dependent upon exogenous shocks and the endogenous determination of prices. An overlapping generations model with monopolistic competition and state dependent pricing is constructed to study the factors that shape the TFPR distribution. The focus is on three key data patterns: (i) countercyclical dispersion of TFPR, (ii) countercyclical dispersion of price changes and (iii) countercyclical frequency of price adjustment. The analysis uncovers two interesting scenarios in which these moments are matched. One arises in the presence of shocks to the dispersion of TFPQ along with a negatively correlated change in the mean of TFPQ. The second arises if the monetary authority responds to shocks to the dispersion of TFPQ by "leaning against the wind". The findings are robust to the introduction of non-CES household preferences. Due to state contingent pricing, the model is nonlinear. Simple correlations mask these nonlinearities of the underlying economy.

"Firms' Sales Expectations and Marginal Propensity to Invest" (with *Andrea Alati, Johannes Fischer, and Maren Froemel*) **[BoE WP]**
How do firms adjust their investment in response to sales shocks and what determines the response? Using a unique firm-level survey, we propose a novel approach to estimate UK firms' marginal propensity to invest (MPI) out of additional income: the forecast error of their sales growth expectations. Investment responds significantly to these sales surprises, with a 1 percentage point unexpected growth in sales translating into a 0.31 percentage point increase in capital expenditure. We find attentive firms to be more responsive, consistent with sales growth surprises providing firms with information about their demand. Sales growth surprises also cause firms to increase their prices, supporting this interpretation. We do not find evidence that these results are driven by financial frictions, uncertainty, or productivity shocks.

"Debt Contracts, Investment, and Monetary Policy"
This paper studies the role of debt contracts on the transmission of monetary policy to firm-level investment and borrowing. Empirically, using information from detailed loan-level data matched with balance sheet data and stock return data, I document that in response to a contractionary monetary shock, asset-based borrowers –firms with more pledgeable assets– experience sharper contraction in borrowing and investment than cash flow-based borrowers –firms with higher profitability. Despite constituting only 15% of the total investment, 64% percent of total investment *response* to monetary policy shocks are initiated by asset-based borrowers. Quantitatively, I set up a heterogeneous firm New Keynesian model with limited enforceability. The findings suggest that the traditional collateral channel explains this heterogeneous sensitivity as cash flow-based borrowers are less vulnerable to asset price fluctuations. Results indicate that debt contract type affects the strength of financial accelerator mechanism and shapes monetary policy transmission.

"Sectoral Volatility and the Investment Channel of Monetary Policy" (with *Thomas Walsh*)
We study how the dispersion of firm-level shocks affects the investment channel of monetary policy. Using firm-level panel data, we construct a measure of volatility based on the dispersion of firm-level productivity shocks and interact high-frequency identified monetary policy shocks with volatility. We document a novel fact: monetary policy has dampened real effects via the investment channel when firm-level TFP shock volatility is high. Given that dispersion rises in recessions, these findings offer further evidence as to why monetary policy is weaker in recessions and emphasize the importance of firm heterogeneity in monetary policy transmission.

"Predicting Financial Constraints: Evidence from the Decision Maker Panel" (with *Nick Bloom, Phil Bunn, Paul Mizen, Greg Thwaites, Ivan Yotzov*)

This paper introduces a novel survey-based approach to predict the likelihood of UK firms being financially constrained. By leveraging unique firm-level survey data with balance sheet information, we develop an empirical model that incorporates key proxies from the literature, including size, leverage, age, liquidity, and dividend-paying status. Our results show that the developed index aligns closely with these established indicators, while highlighting the prominent roles of liquidity and leverage. Importantly, we identify heterogeneity across firm sizes: liquidity emerges as a more significant determinant of financial constraints for smaller firms, while leverage is more critical for larger firms. This suggests that smaller firms may face credit rationing, reducing the relevance of leverage, whereas larger firms, with better access to external finance, are more affected by leverage. The index is validated by its ability to identify firms that experience sharper declines in investment following contractionary monetary policy shocks, thereby capturing a financial accelerator effect.

Work in Progress

"Market Concentration and Monetary Policy Transmission in a Monetary Union" (with *Livia Chitu and Federica Romei*)

"NonLinearities in the Effects of Monetary Policy" (with *Russell Cooper*)

"High Frequency Firm Responsiveness" (with *Nick Bloom, Phil Bunn, Paul Mizen, Greg Thwaites, Ivan Yotzov*)

Teaching

GRADUATE	**TA**, *Macroeconomics, Time Series Econometrics* at the Middle East Technical University
MBA	**TA**, *Macroeconomics for Business* at the Middle East Technical University
UNDERGRADUATE	**TA**, *Introduction to Macroeconomics, Intermediate Macroeconomics I - II, Financial Markets, Introduction to Econometrics I - II* at the Middle East Technical University

Previous Work Experience

2012/11 - 2017/08	***Research/Teaching Assistant,* Middle East Technical University, Department of Economics**, Ankara - Turkey
2015/03 - 2017/08	***Researcher,* Science and Research Council of Turkey**, Ankara - Turkey

Conferences & Invited Seminars

2024	Middle East Technical University; University of Bristol; Bank of England Macro Series; University of Oxford Research Jamboree in Economics
2023	Money, Macro, and Finance Society; European Winter Meeting of Econometric Society
2022	European Winter Meeting of Econometric Society; European Economic Association-Econometric Society European Meeting
2021	European Winter Meeting of Econometric Society; World Finance Congress; Asian Meeting of Econometric Society; Computing in Economic and Finance

Grants

2020/08 - 2021/08	**PhD Completion Grant**, European University Institute, Florence, Italy
2017/09 - 2020/08	**PhD Scholarship**, Italian Ministry of Foreign Affairs, Florence, Italy
2015/03 - 2017/08	**Research Fund #114K957**, Scientific and Technological Research Council, Ankara, Turkey
2012/11 - 2017/08	**Research Fund under Government Contract**, Council of Higher Education Grant, Ankara, Turkey

Language & Software Skills

LANGUAGES	TURKISH *(native)*, ENGLISH *(fluent)*
SOFTWARE	MATLAB, STATA, DYNARE, LaTeX, R (basic)